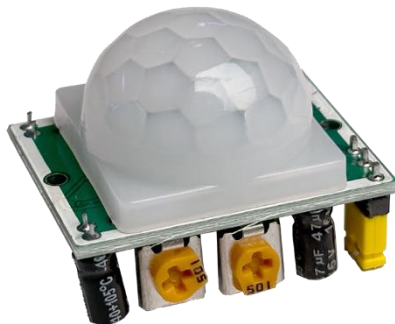


Aim: Motion detection using PIR sensor: Interface a PIR sensor –for motion detection with Arduino and display the result on serial monitor

OBJECTIVES:

1. To understand Arduino UNO board
2. Application of PIR sensor module
3. Detect Motion using PIR Sensor and Arduino Uno

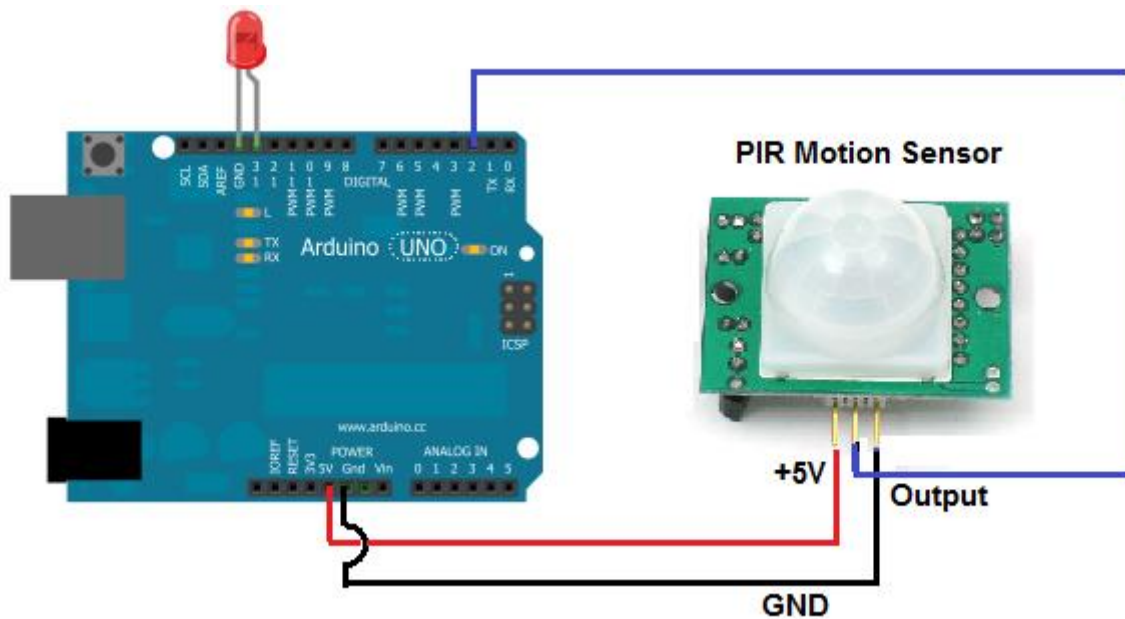
THEORY: PIR sensor The PIR motion sensor is ideal to detect movement. PIR stand for “Passive Infrared”. Basically, the PIR motion sensor measures infrared light from objects in its field of view. So, it can detect motion based on changes in infrared light in the environment. It is ideal to detect if a human has moved in or out of the sensor range.



PIR sensor Pin Configuration:

1. GND – connect to ground
2. OUT – connect to an Arduino digital pin
3. 5V – connect to 5V

Circuit Diagram & Connection:



CODE:

```
const int led = 9; // Led positive terminal to the digital pin 9.
const int sensor = 5; // signal pin of sensor to digital pin 5.
const int state = LOW;
const int val = 0;
void setup() { // Void setup is ran only once after each powerup or reset of the
  Arduino board.
  pinMode(led, OUTPUT); // Led is determined as an output here.
  pinMode(sensor, INPUT); // PIR motion sensor is determined is an input here.
  Serial.begin(9600); } void loop(){ // Void loop is ran over and over and consists
  of the main program.
  val = digitalRead(sensor);
  if (val == HIGH) { digitalWrite(led, HIGH);
  delay(500); // Delay of led is 500 if (state == LOW) { Serial.println(" Motion
  detected ");
  state = HIGH;
  }
  }
  else { digitalWrite(led, LOW);
  delay(500);
  if (state == HIGH){ Serial.println("The action/ motion has stopped");
```

```

state = LOW;
}
}
}

```

RESULT:

The screenshot shows the Arduino IDE 2.3.7 interface. The left sidebar displays the 'BOARDS MANAGER' with various board options. The main editor shows a sketch named 'sketch_jan29a.ino' with the following code:

```

1 // Pin definitions
2 const int pirPin = 2; // PIR sensor ka output pin
3 int sensorState = 0; // Variable motion store karne ke liye
4
5 void setup() {
6   pinMode(pirPin, INPUT); // PIR pin ko input set karein
7   Serial.begin(9600); // Serial communication shuru karein
8
9   Serial.println("Sensor warming up... Please wait.");
10  delay(10000); // PIR sensor ko stabilize hone ke liye 10-20 sec chahiye
11  Serial.println("Sensor Active!");
12
13
14  void loop() {
15    // Sensor se data read karein
16    sensorState = digitalRead(pirPin);
17
18    if (sensorState == HIGH) {
19      // Agar motion detect hua
20      Serial.println("Motion Detected! Someone is there.");
21      delay(500); // Thoda wait karein taaki monitor fill na ho jaye
22    } else {
23      // Agar motion nahi hai
24      Serial.println("No motion.");
25      delay(500);
26    }
27  }
28

```

The 'Serial Monitor' at the bottom shows the output of the sketch:

```

Motion Detected! Someone is there.
Motion Detected! Someone is there.
Motion Detected! Someone is there.
Motion Detected! Someone is there.
Motion Detected! Someone is there.

```

